

SARASWAT ACADEMY

WORKSHEET (CLASS-IX- MATHEMATICS)

CHAPTER- 1: Orienting Yourself: The Use of Coordinates

MCQ'S

- Q1: The point $(-2,-1)$, $(1,0)$, $(4,3)$ and $(1,2)$ are the vertices of a:
(A) Rectangle (B) Parallelogram (C) Square (D) Rhombus
- Q2: The distance of the point $(3, 5)$ from X- axis is:
(A) (B) 3 (C) 5 (D) None of these
- Q3: The x co-ordinate of a point is distance of that point from :
(a) x-axis (b) y-axis (c) Origin (d) None of these
- Q4: Find the distance between $(5,3)$ and $(3,2)$ is:
(a) 5 units (b) $\sqrt{5}$ units (c) $\sqrt{11}$ units (d) None of these
- Q5: If both co-ordinates of any point are negative then that point will lie in :
(A) First quadrant (B) Second quadrant
(C) Thirst quadrant (D) Fourth quadrant

COMPETENCY BASED QUESTIONS

- Q1: Plot the following points on the graph paper.
(i) $A(2, 5)$ (ii) $B(-5, -7)$ (iii) $C(3, -2)$ (iv) $D(0, 5)$ (v) $E(5, 0)$
- Q2: With rectangular axes, plot the points $O(0,0)$, $A(4,0)$ and $C(0,6)$. Find the coordinates of the fourth points B such the OABC forms a rectangle.
- Q3: Find the value of x, if the distance between the points $(x, -1)$ and $(3,2)$ is 5.
- Q4: Given the points $A(1, -8)$, $B(-4, 7)$ and $C(-7, -4)$, show that they lie on a circle K whose center is the origin $O(0, 0)$. What is the radius of circle K?

CASE STUDY

The municipal corporation planned to build a **square-shaped seating area** in a public park. Before laying the tiles, the engineer marked the four corners on a coordinate plane.

The coordinates of the four corners are: **A (1, 3), B (-3, 1), C (-1, -3), D (3, -1)**

The engineering team wanted to verify whether these points form a perfect square. They also needed to calculate the area of the seating area so that the required number of tiles could be ordered.

Using the concepts of **Coordinate Geometry**, answer the following questions.

- Q1. Plot the points A, B, C and D on the coordinate plane. What shape do they appear to form?
- Q2. Find the length of side AB using the distance formula.
- Q3. Find the lengths of BC, CD and DA.
- Q4. Find the lengths of diagonals AC and BD.
- Q5. Can you conclude that ABCD is a square? Give reasons.
- Q6. Find the area of the square.

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